



Equality in an equation

1 Which expression is represented by this bar model? Tick 2 correct answers



☐ $3x + 6$

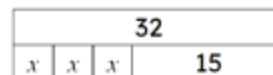
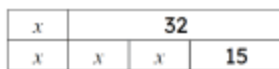
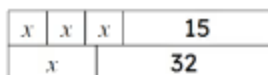
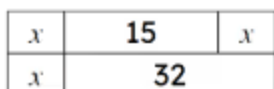
☐ $6x + 6$

☒ $6x + 12$

☒ $2(3x + 6)$

☐ $3(x + 6)$

2 Select the bar model that correctly represents the equation $3x + 15 = x + 32$. Tick 1 correct answer


☐
☐
☒
☐

3 Given that $2.4 + 3.7 = 6.1$, which of these are also true? Tick 2 correct answers

☐ $2.4 + 6.1 = 3.7$

☒ $6.1 - 2.4 = 3.7$

☐ $3.7 - 2.4 = 6.1$

☒ $2.4 = 6.1 - 3.7$

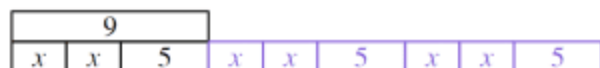


- 4 This bar model represents the initial equation $x + 16 = 27$. What needs to be done to the top bar to maintain equality? Tick **2** correct answers



- ☐ + 2
☐ + 16
☒ + ($x + 16$)
☒ + 27
☐ + x

- 5 This bar model represents the initial equation $2x + 5 = 9$. What needs to be done to the top bar to maintain equality? Tick **2** correct answers



- ☐ + 9
☐ + ($2x + 10$)
☒ + ($4x + 10$)
☐ $\times 2$
☒ $\times 3$



- 6 These 4 bar models represent operations performed on the equation $x + 12 = 19$. Match the bar models labelled A, B, C, D with the equivalent equation they represent. Write the correct letter in each box

A	<table><tr><td>x</td><td>12</td></tr><tr><td>19</td><td></td></tr></table>	x	12	19		<table><tr><td>x</td><td>12</td></tr><tr><td>19</td><td></td></tr></table>	x	12	19		<table><tr><td>x</td><td>12</td></tr><tr><td>19</td><td></td></tr></table>	x	12	19	
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D	<table><tr><td>x</td><td>12</td></tr><tr><td>19</td><td></td></tr></table>	x	12	19		<table><tr><td>x</td><td>x</td><td>x</td><td>12</td></tr><tr><td>19</td><td>19</td><td>19</td><td>19</td></tr></table>	x	x	x	12	19	19	19	19	
x	12														
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x	x	x	12												
19	19	19	19												

a	A
b	B
c	C
d	D

c	$x + 12 + 38 = 19 + 38$
a	$3(x + 12) = 57$
b	$x + 12 + 2x + 24 = 19 + 2x + 24$
d	$4(x + 12) = 76$

